Research Protocol

An Exploratory Neurophenomenological Test of Presence-Complexity Dissociation in Minimal-Content States

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Status: Research proposal for scientific collaboration and methodological review. Not yet conducted or preregistered.

Summary

This protocol proposes a staged, preregistration-intended neurophenomenological investigation of the association between remembered minimal-but-present awareness and concurrent neural signal complexity in contemplative states. The Knower Centred Model of Consciousness (KCM) motivates the question but its ontological claim is not directly tested. The primary proposed neural index is surrogate-normalized EEG Lempel–Ziv complexity (LZc); the primary inferential target is the within-participant association between complexity and the odds of a minimal-presence report relative to no remembered experience, with all four report categories retained in the full analysis. A distinct, adequately powered equivalence analysis could support only a limited claim of practical dissociation. An initial pilot will determine whether target states, usable neural variation, reliable coding and temporally interpretable reports are obtainable. No main-study data collection or preregistration has occurred. This document is offered for critical review and collaborative development, not as a finalized executable study.

A note on scope. Scalp signal complexity is not integrated information (Φ) as defined by Integrated Information Theory (IIT), nor is it a direct measure of consciousness, subjective presence, or neural causal structure. An observed association or dissociation would concern the selected operational measures and retrospective reports only. No outcome alone establishes KCM's ontology or falsifies IIT. Comparisons to particular theories require their own explicit, theory-specific predictions.

1. Background and Rationale

Contemporary consciousness research examines relations among neural organization, conscious level, contents, accessibility, and report. These dimensions should not be conflated. IIT concerns intrinsic causal structure and Φ (Tononi, 2004; Oizumi et al., 2014; Albantakis et al., 2023); scalp EEG signal complexity is a different construct. Signal complexity can also vary with spectral composition, vigilance, preprocessing, and artifacts. Thus the present protocol tests an operational relationship between measured neural complexity and remembered reports, rather than treating low LZc as equivalent to absent consciousness.

KCM distinguishes the Knower, Knowing and Known, and its Modulation Principle proposes that content and access may change without necessarily abolishing subjective presence (Ojha, 2026a–d). This framework motivates a scientifically open question: how do carefully characterized reports of minimal awareness relate to neural signal complexity? The study does not presume that a report of witnessing is evidence of a metaphysically independent Knower, and it does not assume that other theories must predict a fixed complexity–report relationship.

2. Three Distinct Claims and What This Study Can Test

This protocol separates three claims and specifies the limited inference available at each level.

a) **KCM's ontological claim.** The Knower or subjective presence is held to persist independently of the organization or accessibility of experiential content. The claim permits presence without memory or report. The proposed observations cannot directly verify or refute it.

b) **The empirical bridge hypothesis.** Under some operationally identifiable minimal-content conditions, remembered minimal-but-present awareness may remain reportable despite reduced neural signal complexity. This contingent proposition is independently vulnerable to evidence: a sufficiently informative failure to observe the specified report–complexity pattern would weaken this bridge, even though the ontology remains untested. Occurrence at low complexity, weak association, and equivalence are distinct claims; none is inferred automatically from another.

c) **The experimental observation.** Investigators obtain trial-specific phenomenological classifications, uncertainty about continuity, and artifact-controlled neural complexity estimates. These data can estimate report frequencies and associations; they cannot establish uninterrupted subjective presence at every sampled instant.

Inference is asymmetric but not immune to criticism. A null or contrary finding can count against the operational bridge when the design has adequate measurement quality, variation and precision; it cannot establish that subjective presence was absent. A positive finding may support the limited empirical proposition but admits alternatives including residual content, reporting expectations, arousal, memory reconstruction and measurement artifact. All outcomes will be interpreted against prespecified quality and decision criteria, not post hoc metaphysical conclusions.

3. Competing Expectations

	Complexity–report covariance (operational comparator)	KCM-motivated bridge / limited dissociation
General expectation	A positive association between neural LZc and relative odds of minimal-presence	Some minimal-presence reports may occur at low LZc; a sufficiently small association is a separate, testable

	reports	possibility
Low-complexity observations	Low-LZc occurrence is compatible with a positive overall slope	Occurrence alone does not show independence; report frequency and slope must both be estimated
What informative evidence means	A precise positive slope supports covariance for the selected metric and report contrast	Only justified equivalence bounds plus adequate precision support limited practical dissociation
Limits of inference	No operational slope is a universal prediction of IIT or other named theories	Neither outcome establishes KCM ontology or rules out alternative explanations

Scope of comparison. The complexity-covariance account is an explicitly operational comparator: stronger association between the chosen neural metric and the chosen report contrast. It is not attributed universally to IIT, global workspace theories or neuroscience as a whole.

The study distinguishes (i) the occurrence of minimal-presence reports at low complexity, (ii) the direction and magnitude of the report–complexity association, and (iii) whether the association is sufficiently small to meet a prospectively justified equivalence margin. A few low-complexity reports cannot establish statistical independence.

The full four-category report distribution and competing explanations remain central. The study does not select trials for primary analysis by first requiring the outcome category it seeks to predict. Experimental condition labels and prespecified quality criteria define the eligible trial set independently of report outcomes.

4. Research Questions

Primary question. Across valid trials in prespecified contemplative conditions, what is the within-participant association between surrogate-normalized EEG LZc and the odds of a remembered minimal-but-present report relative to a no-remembered-experience report, while retaining uncertain recall and content-rich experience in the full multinomial outcome model?

Primary hypothesis and estimand (H1). The primary estimand is the change in the log-odds of category (iii) versus category (i) per one within-participant standard-deviation increase in the prespecified LZc metric, conditional on study condition and prespecified confounders. A positive slope is consistent with complexity covariance; a near-zero slope, if estimated precisely enough, is compatible with a limited operational dissociation. The sign, magnitude and confidence interval will be reported irrespective of statistical significance.

Secondary questions. Estimate the prevalence of category (iii) in an independently defined low-LZc range, the complete four-category distribution, and between-person heterogeneity. A separate equivalence test may ask whether the primary slope falls entirely within a justified smallest-effect interval. The equivalence bounds, confidence level, minimum usable complexity range and required precision must be set with a statistician using pilot-independent rationale before main-study preregistration; failure to reject a null association is not equivalence.

Operational alternatives. A reliably positive slope supports covariance for these measures; a sufficiently precise slope inside prospectively justified equivalence bounds supports a limited absence of meaningful association; a negative slope is a distinct result requiring explanation. Sparse categories, uncertain recall, insufficient neural variation or wide intervals yield inconclusive results, not automatic support for KCM.

All directional expectations, contrasts, quality gates and interpretive rules will be locked before the main-study dataset is acquired. The pilot is exploratory and must be labeled as such; pilot-derived design changes will be documented and evaluated on independent main-study data.

5. Study Design

A staged within-subject design comprises a feasibility pilot followed, only if prespecified criteria are met, by a separately preregistered main contemplative study. Three possible clinical or altered-state extensions are independent future studies, not pooled arms of the primary analysis.

Primary contemplative study. Proposed conditions are (i) resting wakefulness, (ii) focused attention, (iii) open monitoring, and (iv) absorption/nondual practice. Approximately 8–12 trials of 2–4 minutes per condition are an initial engineering proposal only. The pilot will compare feasible trial durations and self-paced versus externally cued termination, document practice disruption and fatigue, and select one standardized main-study schedule. Condition eligibility will be defined by assigned instruction and objective recording-quality criteria, not by the eventual minimal-presence report.

Pilot feasibility and go/no-go gates. A separate pilot will quantify (a) successful completion and artifact-free neural recording, (b) adequate within-person LZc spread including independently defined low-complexity observations, (c) ability to enter the intended practice condition without excessive disruption, (d) timely and interpretable reports, (e) four-category coding reliability, and (f) sufficient category frequencies for planned inference. The pilot will also check whether absorption decreases, increases or leaves LZc unchanged; no decrease is assumed. Numerical feasibility thresholds and a stop/redesign decision must be specified in a pilot plan before pilot data are examined. Failure of any essential gate triggers redesign or termination rather than a confirmatory test.

Secondary extension 1: Anaesthesia induction and emergence. In a clinical or research setting with appropriate oversight, using the host site's existing anaesthesia research protocols, with added serial probing and structured interview at emergence.

Secondary extension 2: N3 slow-wave sleep with scheduled awakenings. Using the host site's existing sleep research protocols, with added structured interview immediately following each scheduled awakening.

Secondary extension 3: Psychedelic-induced content-minimal states. Dependent on a specialized collaborating site with existing psychedelic research infrastructure and ethics approval, informed by recent work showing that certain psychedelic states can preserve wakefulness while dissolving most or all reportable content (Timmermann et al., 2025).

These three secondary extensions are presented separately, not as a single combined arm, because their ethics approval pathways, measurement constraints, and reportability characteristics differ substantially from one another and from the primary arm, and each would need its own protocol refinement with an appropriately specialized collaborating site.

6. Participants

Primary arm. Recruit practitioners with documented experience in the relevant contemplative practice and demonstrated ability, during screening, to undertake the assigned conditions; lifetime hours (initially 500–1,000 as a recruitment guide) are not a substitute for demonstrated capability. Record tradition, training, years/hours, prior experiences, and pre-existing beliefs about enduring witnessing or awareness. The initial n = 20–30 is a feasibility planning figure, not a justified main-study sample size. Main-study n and usable trials will be set by simulation-based precision/power calculations that account for clustering and rare report categories.

Comparison and secondary populations. Where feasible, recruit practitioners from differing traditions or belief profiles and assess reporting-expectation effects; any nonpractitioner comparison must not be assumed able to achieve equivalent states. Anaesthesia, sleep and psychedelic studies require separate eligibility, consent, clinical oversight, ethics and preregistration; none contributes to the primary estimand here.

Recruitment and safeguards. Recruitment may occur through contemplative centers and university research programmes. Record selection pathways and nonparticipation where feasible; avoid recruiting only participants who already endorse a persistent witness. Prespecify relevant safety exclusions, accommodation needs, withdrawal rights, and compensation with the host ethics committee.

7. Measures

Third-person measures. The proposed primary neural index is surrogate-normalized Lempel–Ziv complexity from scalp EEG, computed on fixed-length, artifact-controlled, prespecified windows. EEG is the primary platform for operational consistency; MEG requires a separate validated processing specification. This index measures signal properties, not Φ, the Knower, or consciousness itself.

- **Primary EEG LZc.** Before main-study preregistration, lock the electrode montage and reference, sampling frequency, filtering and artifact-removal pipeline, window length and placement, signal discretization, LZ algorithm, surrogate construction and normalization, aggregation across channels/windows, minimum usable data, and software versions. Define low-LZc ranges from baseline or pilot-independent distributions without consulting outcome labels. Report sensitivity to spectral power, vigilance, EMG/EOG and reasonable processing alternatives. The choice of spontaneous EEG LZc and shuffled-sequence normalization is informed by Schartner et al. (2015); application to contemplative states and the precise pipeline remain to be validated in the proposed pilot.

- **PCI/TMS–EEG.** Perturbational Complexity Index (Casali et al., 2013) is not interchangeable with resting EEG LZc. It may be examined only as a separately specified exploratory extension at a site with appropriate TMS–EEG expertise, ethics approval and compatible procedures; it is not a confirmatory outcome of this protocol.

- **Secondary neural and physiological measures.** Prespecify spectral power and vigilance/drowsiness indices, ocular and muscle artifacts, movement, and other available arousal measures as quality indicators or covariates. Connectivity and alternative complexity metrics are exploratory, with multiplicity and interpretation clearly separated from the primary analysis.

First-person measures. Following each trial, collect a brief non-leading open-ended description before any categorical question. Use a standardized neutral probe about what, if anything, is remembered from the specified interval; then record perceived continuity (continuous/intermittent/uncertain), confidence, content richness, and subsequent interpretation as separate fields. A stratified, prespecified subset receives trained micro-phenomenological interviewing (Petitmengin, 2006), blind to neural measurements.

(i) no remembered experience: no retrievable experiential memory for the target interval; this does not establish absent experience;

(ii) uncertain or indeterminate recall: the participant cannot reliably decide what, if anything, was experienced or remembered;

(iii) remembered minimal-but-present awareness: a first-person report of minimal experiential presence with little described conceptual, perceptual or narrative content, supported by the open-ended account rather than by endorsement of "witness" terminology;

(iv) remembered content-rich experience: a report of identifiable conceptual, perceptual, bodily, emotional or narrative content beyond the minimal-content coding threshold.

Phenomenological validity and expectation controls. Interviewers use a scripted, neutral sequence and do not introduce KCM vocabulary before spontaneous reporting. Record participants' contemplative tradition and baseline beliefs separately. Independent raters blinded to neural results,

assigned condition where practicable, and study hypothesis code descriptions classify de-identified narratives. A second adversarial/independent reviewer will assess borderline minimal-presence classifications and possible learned interpretations.

Codebook development. Pilot interviews will be used to refine operational examples, exclusion cases and adjudication rules. Freeze the codebook before main-study data collection. Keep raw description, immediate category, continuity/confidence, and later metaphysical interpretation as distinct variables. Predefine double-coding, disagreements, adjudication and audit trails; do not silently relabel uncertain reports.

Reliability. Assess nominal Krippendorff's alpha with uncertainty intervals and category-specific agreement; target $\alpha \geq 0.80$ before the main study. Values near 0.67 may justify continued pilot exploration but do not meet the main-study gate without a documented remedial plan. Category (i) concerns memory, never demonstrated absence of experience. Low prevalence and prevalence-related reliability distortions will be reported.

8. Procedure

1. Screen and consent; document practice capability, baseline beliefs, health/safety eligibility, and understanding of neutral report instructions.

2. Collect standardized resting baselines and physiological vigilance indicators. Use baseline data to characterize complexity distributions without outcome-informed threshold selection.

3. Conduct randomized or counterbalanced condition blocks with adequate transition/rest periods. Pilot both self-paced and unobtrusive cued exits, assess disruption, then select a single main-study procedure. Time-stamp all events and report windows.

4. Prespecify neural analysis windows and exclude onset/offset, motor preparation and cue-related buffers. Quantify residual movement and ocular/muscle artifacts; use only windows meeting predefined quality criteria.

5. Immediately obtain an open-ended report followed by neutral category/continuity/confidence questions. Conduct deeper interviews on a prespecified stratified subset; avoid substituting delayed interview interpretations for immediate reports.

6. Process neural data and code reports in independent blinded streams; freeze quality decisions and labels before linking datasets. Preserve anonymized raw-to-analysis provenance and software versions.

Temporal alignment. Define the target reporting interval and neural summary window before the main study. Ask whether the described state occupied the full interval, part of it, or an uncertain duration. A trial-level report cannot identify the exact neural epoch of a momentary experience. Analyses of purportedly continuous reports and shorter aligned windows are sensitivity analyses, not evidence of instantaneous matching.

Selection and circularity safeguards. The primary dataset includes all eligible trials from prespecified conditions meeting neural quality criteria, irrespective of subsequent report category. Do not select "minimal-content trials" using category (iii) and then use that same category as the primary outcome. If a separate minimal-content screen is used, it must be defined independently and its impact reported transparently.

Secondary extensions, if pursued, require independent timing, reportability and safety procedures appropriate to the setting; no extrapolation of the primary-arm temporal claims is assumed.

9. Analysis Plan

Primary estimand and population. Among all eligible, artifact-controlled trials in assigned contemplative conditions, estimate the conditional within-person log-odds slope for category (iii) versus (i) per one within-person standard-deviation increase in surrogate-normalized EEG LZc. The four nominal categories remain in the full outcome model; categories (ii) and (iv) are not discarded or recoded as absence. Define the target EEG summary window, participant-centering method and fixed effects before main-study acquisition.

Primary statistical model. Fit a hierarchical multinomial regression with report category as nominal outcome, participant-level random intercepts (and a random LZc slope if supported), within-participant centered LZc, condition, prespecified vigilance/spectral covariates and trial order. Report slope estimates, uncertainty and category probabilities. The statistician will predefine an estimable fallback (e.g., penalized/Bayesian partial-pooling model) for sparse cells or nonconvergence, rather than selecting whichever model yields a desired result.

Equivalence and feasibility. A separate two-one-sided-tests or interval-based equivalence assessment may be preregistered only after a scientifically justified smallest-effect margin and adequate precision are established. A nonsignificant slope alone is not evidence of dissociation. Pilot-based simulations must account for participant clustering, category prevalence, measurement error, missingness and the actual LZc range; sample size will be revised accordingly. Low-complexity occurrence frequencies are secondary descriptive estimates with intervals.

Analysis integrity. Prespecify baseline-derived low-LZc definitions, quality exclusions, handling of uncertain recall, incomplete reports and missing neural data, multiple-comparison control for secondary tests, and sensitivity to alternative windows, preprocessing and belief/arousal covariates. Assess whether results persist across participants and conditions. Preserve reproducible code, versioned processing, de-identified data where ethically permissible, and a deviation log.

Registration sequence. The pilot is exploratory and governed by a prospective feasibility plan. Only after pilot evaluation and design freeze will the independent main study be preregistered on OSF or equivalent, before its first participant is enrolled or outcome data are collected. Register hypotheses, primary estimand, metric pipeline, codebook, quality gates, exclusions, sample-size rationale, statistical model, equivalence bounds if applicable, and all decision rules. Report negative, inconclusive and infeasible outcomes.

10. Interpretation of Outcomes

Interpretation is restricted to operational measures and depends on passing feasibility and measurement-quality gates. The ontology of the Knower remains outside the direct evidential reach of this study.

- Evidence for positive covariance: a sufficiently precise positive primary slope indicates greater odds of remembered minimal presence relative to no remembered experience as EEG LZc increases, conditional on the model and its assumptions.

- Evidence for limited practical dissociation: only a prospectively justified and adequately precise equivalence result, with credible report coding, usable LZc range and robustness checks, supports a small association for these measures. Occurrence of category (iii) in a few low-LZc trials is descriptive, not proof of independence.

- Negative, unexpected, inconclusive or infeasible outcomes: a negative slope warrants independent interpretation; wide intervals, sparse categories or high uncertain-recall rates are inconclusive; inadequate state production, neural variation or data quality are feasibility failures. No category is reinterpreted after seeing results to rescue a preferred theory.

All findings will be published or otherwise reported regardless of direction. No finding by itself establishes the persistence of a metaphysical Knower, disproves brain dependence, or falsifies IIT; specific theory comparisons require explicit additional assumptions and predictions.

11. Limitations

- Ontological underdetermination. The study tests a contingent reportability bridge, not the existence or independence of the Knower. Other theories can accommodate similar findings.

- Construct and metric limitations. EEG LZc depends on spectral content, vigilance, preprocessing, montage and artifacts; it is not Φ or a direct consciousness meter. A low LZc value is relative to a defined reference, not a universal neural floor.

- Retrospective and temporal limitations. Memory reconstruction, intermittent states, uncertain recall and imperfect neural–phenomenological alignment prevent moment-by-moment inference. Minimal content is not necessarily zero content.

- Expectations and reporting. Meditation training, demand characteristics, vocabulary, interviewer influence and philosophical commitments can shape reports despite neutral interviewing and blinded coding.

- Selection and generalizability. Expertise-based recruitment and condition differences can introduce selection effects; comparison groups may differ in ability to produce the target state. A within-person association is not a causal effect of complexity.

- Feasibility, burden and ethics. Brief trials, external cues, button presses, fatigue and repeated interviews may alter the target experience. Independent ethics review, informed consent, monitoring, stopping rules and privacy safeguards are required. Clinical, sleep and psychedelic extensions require distinct approvals.

- Prior work and novelty. Related EEG and phenomenological studies, including Timmermann et al. (2025), precede this proposal. The intended contribution is an explicitly constrained, reproducible test of a particular report–complexity relationship, not priority or decisive theory falsification.

12. What This Protocol Needs From a Collaborating Site

- EEG laboratory with documented preprocessing and artifact-control expertise; contemplative/neurophenomenological researchers trained in neutral interviewing and independent coding.

- A feasibility-pilot plan specifying state-production checks, usable LZc range, trial/cue options, report timing, codebook reliability and explicit go/no-go gates before pilot analysis.

- Institutional ethics oversight, participant safety procedures, data governance, secure storage and a protocol for adverse events or withdrawal.

- A biostatistician to define the primary estimand, simulations, model/fallback, missingness rules and any defensible equivalence margin; a blinded independent rater to audit expectation-related classifications.

- A reproducible EEG processing workflow, preregistration-ready codebook, anonymization plan and appropriate consultation before making theory-specific comparisons to IIT or other frameworks.

- Agreement to preregister the main study only after pilot-driven design freeze and to disseminate all outcomes, including failures of feasibility, negative findings and inconclusive estimates.

This version is a rigorous proposal for co-development, not a completed study or an approved preregistration. The remaining open numerical decisions—pilot gates, acquisition parameters, sample size, smallest-effect margin and site-specific safeguards—are intentionally identified rather than invented. A collaborating laboratory must finalize and prospectively lock them before confirmatory testing.

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